User Manual



V 1.0

This user manual explains the analytical background, input requirements, parameter settings, operating steps, exports, and interpretation of results for all four EvoTempo modules. Use it alongside the Quick Guide, which provides a concise route through the main workflow.

This manual corresponds to EvoTempo v1.0. Software and package versions may evolve; always verify parameter names against the running interface.

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1. Overview

1.1 Introduction

EvoTempo is an interactive R/Shiny toolkit for evaluating temporal signal in measurably evolving populations. Designed for viral phylodynamicists and evolutionary biologists, it integrates complementary approaches for examining temporal – genetic association, exploring root-to-tip temporal structure, preparing and evaluating date-randomization tests, and comparing models using Bayesian evaluation of temporal signal.

EvoTempo provides these analyses within a unified graphical interface, reducing the need for complex command-line workflows while supporting reproducible analysis and publication-ready visualization.

1.2 Modules in EvoTempo

EvoTempo integrates four functional modules within one graphical interface. Module 1 provides the tree and sampling dates required by Module 2. Modules 3 and 4 also support standalone use with externally prepared XML configurations, MCMC logs, or marginal-likelihood estimates, as appropriate:

- Module 1: Data Parsing & Mantel Test
- Module 2: Interactive Root-to-Tip (RTT) Regression
- Module 3: Date-Randomization Test (DRT)
- Module 4: Bayesian Evaluation of Temporal Signal (BETS)

1.3 Recommended Workflow

Sequential workflow: prepare and validate the data in Module 1, run the Mantel test if required, and explore root-to-tip patterns in Module 2. For DRT, supply a configured BEAST XML in Module 3; if using CSV-based clustered permutation, export the cluster metadata from Module 1 and upload it explicitly into Module 3. Execute BEAST analyses externally before evaluating logs or marginal-likelihood summaries in Modules 3 and 4.

Standalone workflow: open Module 3 with a prepared XML template or a set of original and randomized logs, or open Module 4 with externally estimated log marginal likelihoods. Module 2 requires data loaded through Module 1; use **Send to RTT (Skip Mantel Test)** when Mantel testing is not needed.

The specific workflow is shown in **Figure 1**.

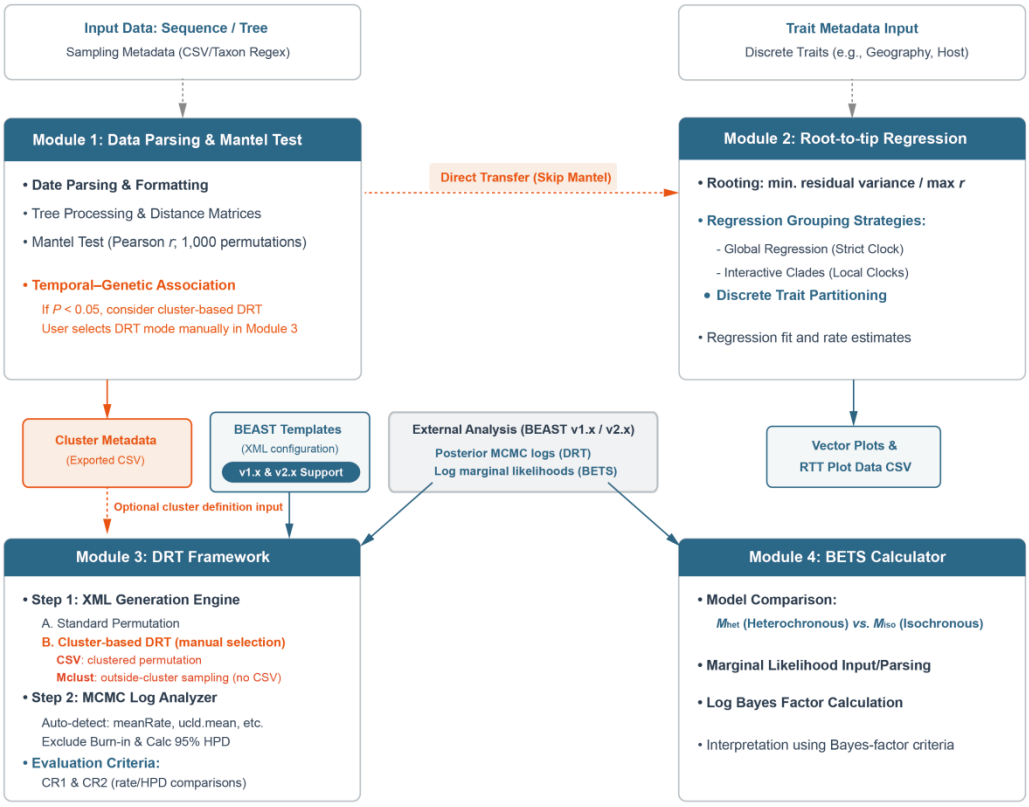


Figure 1 Overview of the EvoTempo workflow. Module 1 supplies the tree and dates used by Module 2. Module 3 accepts an independent BEAST XML template and optionally uses uploaded cluster metadata or automatic date clustering. BEAST execution and marginal-likelihood estimation take place outside EvoTempo.

2. Data Preparation & Prerequisites

Before initiating any analysis, ensure your input files are formatted correctly. EvoTempo relies on standardized file formats common in phylodynamic workflows (e.g., BEAST, IQ-TREE, RAxML).

- **Phylogenetic tree (.tre, .nwk, .tree, .nex)**: provide a valid Newick or NEXUS tree with branch lengths. Rooted and unrooted trees can be used; Module 2 optimizes rooting for RTT analysis. Check the branch-length units and remove invalid or negative lengths before upload. When a file contains multiple trees, Module 1 uses the first.
- **Sequence alignment (.fasta, .fas)**: provide an aligned nucleotide dataset with consistent, unique sequence identifiers. In Module 1, upload either an alignment or a tree. When an alignment is supplied, EvoTempo calculates TN93 distances with pairwise deletion and constructs a neighbor-joining tree; a separate tree upload is not required.
- **Sampling dates**: use a consistent identifier format, such as Taxon_Country_2020-05-12 or Taxon|2020.35, and configure both date extraction and format conversion in Module 1. Alternatively, provide a metadata CSV with a taxon identifier column and a date column. Match metadata identifiers to the sequence or tree labels and inspect the decimal-year preview before analysis.
- **BEAST XML (.xml)**: required for the XML Generator in Module 3. The parser recognizes BEAST 1.x date elements (<date value="..."/>) and BEAST 2.x date or date-forward trait elements. It uses line-based parsing, so the relevant trait attributes and date-value list must be on the same input line for the supported BEAST 2.x format. Inspect templates with other layouts or date uncertainty before use.
- **MCMC logs and marginal-likelihood summaries**: the DRT Log Analyzer reads the original and randomized BEAST .log trace files containing the selected rate parameter. The BETS calculator instead requires final log marginal likelihood

estimates, entered manually or extracted from .log, .txt, or .out summaries. An ordinary posterior trace is not a marginal-likelihood estimate.

Prepare only the inputs needed for the selected analysis. XML generation, posterior-log evaluation, and BETS model comparison have separate input requirements.

Bundled examples are located under inst/Mantel_Test, inst/RTT, inst/DRT, and inst/BETS relative to the application folder. Use these files to inspect the expected formats before preparing your own inputs.

Table 1 summarizes the input data, key configuration options, and outputs of the four EvoTempo modules.

Table 1 Input data, key configuration options, and outputs of the four EvoTempo modules.

Module	Required input(s)	Key configuration options	Main output(s)	Analytical purpose
Module 1: Data processing and Mantel test	Phylogenetic tree(.nwk, .tree, .nex) or sequence alignment (.fas, .fasta) ; optional sampling date metadata (.csv)	Delimiter or regex-based date parsing; interactive decimal-year table editing; Mantel permutation count (default: 1,000)	Parsed tree and date vector; temporal and genetic distance matrices; Mantel *r* and *P*-value; phylo-temporal cluster tree (PDF) and cluster metadata (.csv)	Parse sampling dates and examine temporal–genetic association
Module 2: Interactive RTT regression	Phylogenetic tree and corresponding sampling dates (passed from Module 1); optional discrete trait metadata (.csv)	Rooting strategy (nve-rms or max r); grouping mode (Global Clock, Interactive Clades / Local Clocks, or Discrete Trait Partitioning); permutation test (1,000 replicates)	Color-synchronized interactive phylogeny and RTT regression plot; group-specific Rate, *x*-intercepts, and R^2 summaries; grouped data (.csv) and vector plots (PDF)	Explore clock-likeness, lineage-specific rate heterogeneity, and root-to-tip temporal structure

Module 3: Date-randomization test (DRT)	BEAST v1.x/v2.x XML template; optional cluster metadata (.csv/.tsv); original and randomized MCMC .log files	Randomization mode (Standard *vs.* Cluster-based); clustering and randomization strategy (CSV-based clustered permutation vs. Mclust-based outside-cluster sampling); requested replicate count (5–20); Mclust model (E/V), maximum G and optional date rounding; burn-in percentage and rate parameter	ZIP archive of date-randomized BEAST XMLs and a randomization-strategy record; 95% HPD interval summaries; CR1 and CR2 validation cards; interactive and PDF 95% HPD forest plot	Evaluate temporal signal by comparing posterior rate estimates against a date-randomized null distribution
Module 4: Bayesian evaluation of temporal signal (BETS)	Marginal likelihood output logs (GSS, PS, or SS) or manual log-marginal likelihood values for competing models	Competing model specification (M_{het} vs. M_{iso}); automated log marginal likelihood text extraction or manual entry	Log marginal likelihood estimates; log Bayes Factor; Kass & Raftery evidence classification; dumbbell comparison plot (PDF)	Formally evaluate temporal signal via Bayesian model selection between heterochronous and isochronous models

Abbreviations: RTT, root-to-tip; tMRCA, time to the most recent common ancestor; DRT, date-randomization test; HPD, highest posterior density; BETS, Bayesian evaluation of temporal signal; MCMC, Markov chain Monte Carlo; GSS, generalized stepping-stone sampling; PS, path sampling; SS, stepping-stone sampling; M_{het}, heterochronous model; M_{iso}, isochronous model.

3. Installation & Dependencies

EvoTempo is an R/Shiny application. All dependencies must be installed before launching the software.

3.1 System Requirements

- **R ≥ 4.1.0**
- **RStudio ≥ 2022.02** (recommended, not required)
- **Operating system**: Windows, macOS, or a mainstream Linux distribution
- **Web browser**: Chrome 90+, Edge 90+, Firefox 88+, or Safari 14+

3.2 Required R Packages

Install the dependencies listed in the software DESCRIPTION file before launching EvoTempo. The CRAN installation command below includes the main external dependencies; ggtree is installed separately through Bioconductor. The optional mclust package is needed only for automatic date clustering in Module 3 (Section 3.3).

Install CRAN packages in R:

```
install.packages(c(
  "ade4", "ape", "bslib", "coda", "crosstalk",
  "dplyr", "DT", "future", "ggplot2", "lubridate",
  "plotly", "promises", "seqinr",
  "shiny", "shinyjs", "stringr", "zip"
))
```

Note: `tools` and `utils` are supplied with R and do not require separate installation. Use an R/Bioconductor combination compatible with the package versions being installed.

Install Bioconductor packages (`ggtree`):

`ggtree` is distributed through Bioconductor, not CRAN. Install it using BiocManager:

```
if (!requireNamespace("BiocManager", quietly = TRUE))
    install.packages("BiocManager")


BiocManager::install("ggtree")
```

Advanced alternative: ggtree can also be installed from its GitHub repository. Check compatibility with your R and Bioconductor versions before using a development version:

```
install.packages("remotes")
remotes::install_github("YuLab-SMU/ggtree")
```

3.3 Optional Packages

- mclust: required only for **Auto-cluster (Mclust on dates)** in Module 3. Install it with install.packages("mclust") when using that strategy. **Standard DRT** and **User-defined clusters (CSV)** do not require mclust.

3.4 External Software (Optional but Recommended)

- **BEAST 1.10.x or BEAST 2.x**: required to execute the corresponding original and randomized XML analyses and to produce posterior logs or marginal-likelihood summaries. EvoTempo prepares inputs and summarizes outputs; it does not execute BEAST.
- **Tracer v1.7.x**: recommended for inspecting posterior traces, convergence, effective sample sizes, and burn-in before DRT log evaluation. These diagnostics are not calculated by EvoTempo.

3.5 Launching EvoTempo

Clone or download the repository. In R, set the working directory to the application folder containing app.R, global.R, and the R subdirectory, then launch the application:

```
setwd("path/to/EvoTempo")
shiny::runApp()
```

Alternatively, open app.R in RStudio and click Run App.

4. Module 1: Data Input & Mantel Test

Module 1 prepares the tree and sampling dates for RTT analysis and tests the association between temporal and genetic distances. A significant Mantel result can motivate closer examination of sampling structure, but does not by itself establish confounding or validate a molecular-clock signal.

4.1 Uploading Data

Navigate to the **Data & Mantel Test** tab. Under **Sequence / Tree File**, upload either a phylogenetic tree or a nucleotide alignment and check the **Format** selection. Under "**Where are the dates?**", choose **Parse from Taxon ID** or **Upload Metadata (CSV)**. The preview table updates automatically as the date settings change.

For metadata-based dates, choose **Upload Metadata (CSV)**, upload the comma-separated file, and select the taxon identifier and date columns. Use identifiers that match the uploaded sequence/tree labels. Inspect every parsed date in the preview; double-click a **Decimal Year** cell to correct it. Resolve all NA values, which are highlighted in red, before running or transferring an analysis.

For identifier-based dates, choose **Delimiter & Order** or **Regular Expression** under **Extraction Method**. With **Delimiter & Order**, specify the literal separator and use first, last, or a numeric token index; for example, the last token of Taxon_Country_2020.35 is 2020.35. Check the **Extracted String** column because the parser can fall back to another date-like token when the selected token does not look like a date.

Under **Format Transformation**, select **Number (Decimal Year)** for numeric years, or **Calendar Date** for calendar strings and specify **Date Format**, for example yyyy-MM-dd. Check ambiguous day/month formats explicitly. **Enable 2-digit year correction** only for dates that require it: by default, values below 16 receive 2000 and other values below 100 receive 1900. The preview rounds decimal years to four places, and those preview values are used in the subsequent analysis.

To proceed directly to Module 2, click **Send to RTT (Skip Mantel Test)** after checking the preview. This transfers the current tree and decimal years without a Mantel result or cluster assignments. If dates are edited later, run the Mantel test or transfer the data again, then recompute RTT in Module 2.

4.2 Running the Mantel Test

The Mantel test compares pairwise genetic distances with absolute differences in sampling dates. A significant association can reflect accumulated evolution, structured sampling, or both. Interpret it alongside the sampling design and phylogeny when choosing subsequent analyses.

Step 1: Set Mantel Permutations (default 1,000; minimum 100), check the decimal years, and click Run Mantel Test.

Step 2: EvoTempo computes Pearson's r between the finite pairwise genetic and temporal distances. Tree inputs use patristic distances; alignment inputs use TN93 distances. It permutes the dates and reports the proportion of correlations at least as large as the observed value. The current calculation has no plus-one adjustment*:* $P = 0$ means that no sampled permutation reached the observed value.

Step 3: Interpret the result. $P < 0.05$ is displayed as **Significant Temporal–Genetic Association**, and $P \geq 0.05$ as **No Significant Temporal–Genetic Association**. Neither result alone confirms or excludes temporal signal or all forms of confounding. For a significant result, EvoTempo groups monophyletic clades sharing an identical parsed date, with remaining tips treated as individual clusters, and displays the cluster tree (**Figure 2**). Use **Download Cluster Metadata (.csv)** to export the Taxon_ID and Cluster assignments for a subsequent CSV-based DRT when appropriate.

Note: the Mantel result does not automatically toggle **Enable Cluster-based DRT** in Module 3 or upload its cluster metadata. Select the strategy explicitly using the sampling design and diagnostics. A non-significant Mantel result does not require use of **Standard DRT** if other evidence calls for a different analysis.

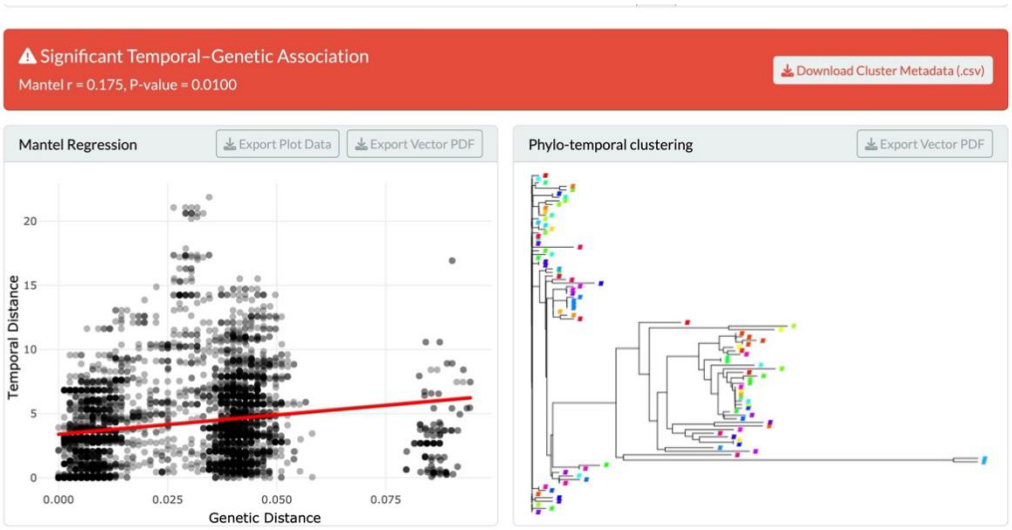


Figure 2 Mantel-test and clustering output from Module 1. The tree displays the phylo-temporal clusters generated following a significant temporal–genetic association; it is a diagnostic grouping rather than proof that confounding has been established.

5. Module 2: Interactive Root-to-Tip (RTT) Regression

This module explores clock-likeness by regressing root-to-tip genetic distance on sampling date. First supply the tree and dates through Module 1, using **Run Mantel Test** or **Send to RTT (Skip Mantel Test)**. Module 2 has no independent tree/date upload. The displayed Rate is an exploratory regression slope; its units are substitutions/site/year only when the branch lengths and dates use those units. The displayed tMRCA is the x-intercept approximation, not a Bayesian node-age estimate.

5.1 Rooting Strategy

Choose a **Rooting Strategy** and click **Compute RTT**. EvoTempo searches alternative root positions and optimizes the root-edge split using the full dataset. The selected root is shared by all regression groups; groups are not independently re-rooted.

- **Minimize Residual Variance (nve-rms)**: the default objective selects the root that minimizes the residual sum of squares divided by the regression residual degrees of freedom. This is an exploratory fitting criterion and does not establish neutrality or clock validity.
- **Maximize Correlation (*r*)**: selects the root with the highest signed Pearson correlation between root-to-tip distance and sampling date. Rooting choices can affect the slope and x-intercept; report the criterion used when presenting RTT results.

5.2 Regression Grouping Strategy

Choose **Regression Grouping Strategy** to inspect the full dataset or selected groups. All three views use exploratory regressions on distances from the same optimized root. The labels **Strict Clock** and **Local Clocks** describe interface views; this module does not fit phylogenetic clock models.

- **Global Regression (Strict Clock)**: fits one regression line to the full dataset and provides a baseline for comparing the grouped views (**Figure 3**). Examine the

scatter, slope, r and R^2 together; a positive slope alone is not formal validation of temporal signal.

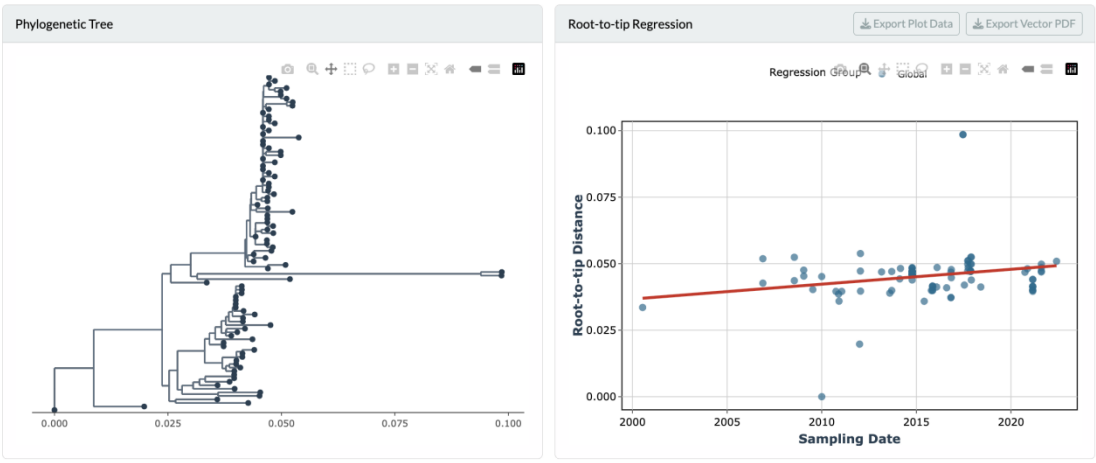


Figure 3 Global Regression in Module 2.

- **Interactive Clades (Local Clocks)**: after computing RTT, click internal nodes on the interactive tree to assign separate regression lines to selected clades. Assigned groups are color-coded (**Figure 4**). **Clear Local Clocks** resets the assignments; later selections can overwrite earlier assignments for overlapping tips. Use slope differences to identify patterns for further investigation.

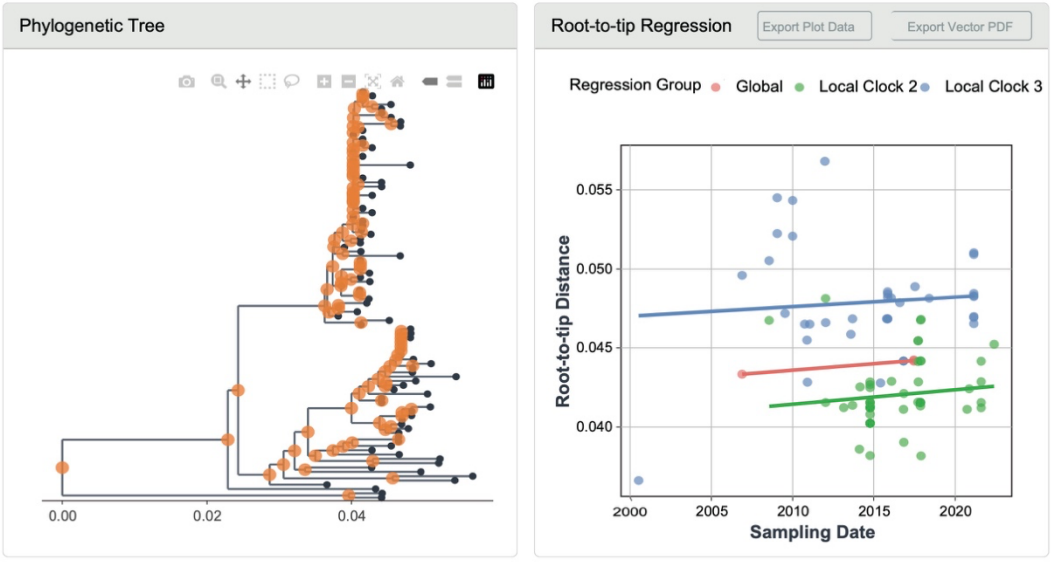


Figure 4 Interactive RTT regression in Module 2. Selected clades allow exploratory comparisons of regression slopes and temporal associations.

- **Discrete Trait Partitioning**: upload metadata containing a taxon identifier column and a categorical trait such as location or host. The reader accepts comma-, semicolon-, and tab-delimited text. Select **Taxon ID Column** and **Grouping Trait**, then click **Apply Trait Grouping (Figure 5)**. Check the reported matching before interpreting groups; blank or unmatched assignments are labeled **Unknown**.

Rate, tMRCA and R^2 are descriptive group summaries. The summary requires more than two observations, more than one distinct date, and a finite slope whose absolute value exceeds the numerical tolerance. Small or uninformative groups can display N/A. The tMRCA value is the negative y-intercept divided by the slope and can be unstable when the slope is close to zero.

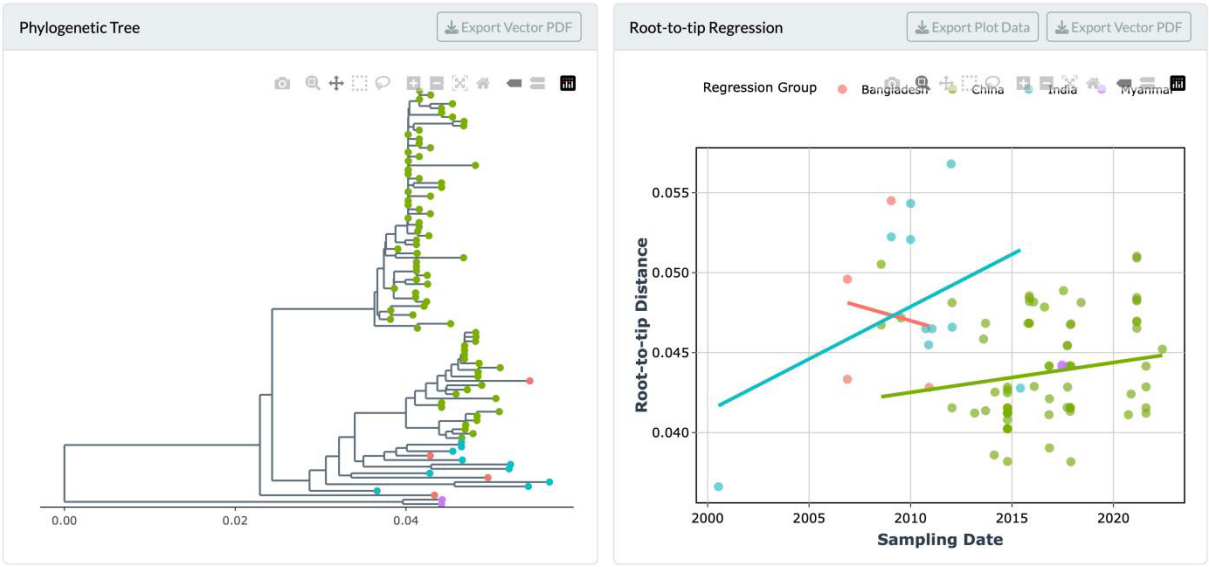


Figure 5 Discrete Trait Partitioning in Module 2.

5.3 Permutation Test

Select **Run Permutation Test (calculate *P*-value)** before clicking **Compute RTT** to shuffle tip dates 1,000 times. The P value is the fraction of absolute randomized correlations at least as large as the absolute observed correlation, without a plus-one adjustment. Distances are held fixed at the optimized global root; the root is not re-optimized for each shuffle. This is an exploratory global diagnostic, not a separate test for each group or a formal validation of phylogenetically independent observations.

5.4 Exporting Results

Both panels support vector export:

- **Phylogenetic Tree** panel → **Export Vector PDF.**
- **Root-to-tip Regression** panel → **Export Plot Data** (CSV) and **Export Vector PDF**. The CSV contains Taxon, Date, Distance, Residuals, and Group. Residuals refer to the global fit, even when grouping is displayed.

6. Module 3: Date Randomization Test (DRT) Engine

DRT evaluates whether rate estimates obtained with the original sampling dates are distinguishable from those obtained under a specified date-randomization design. EvoTempo provides XML preparation and posterior-log evaluation as separate steps. Execute the original and randomized analyses externally in BEAST using comparable model settings, then inspect convergence before evaluating the logs.

6.1 XML Generator (Permutation)

Step 1: Under **Select Step**, choose **Step 1: XML Generator** and upload a configured BEAST XML template (**Figure 6**). **Standard DRT** is the default. To use a cluster-based strategy, select **Enable Cluster-based DRT** and choose an option under **DRT strategy**:

- **Standard DRT**: permutes the original sampling dates across individual taxa without replacement, preserving the frequency of each date. Use the sampling design and preliminary diagnostics to judge whether this randomization is appropriate; the choice is not determined by a Mantel P-value threshold alone.
- **User-defined clusters (CSV)**: supplies cluster assignments for clustered permutation adapted from Murray et al. (2016). Upload a CSV or TSV file, for example the cluster metadata exported from Module 1, and select **Taxon Name Column** and **Cluster/Location Column**. The latter control selects the cluster identifiers; a location label should be used only if it represents the intended grouping.

CSV requirements: include a non-empty, unique taxon name for every row and a non-missing cluster identifier for every XML taxon. Names are matched to the XML, so CSV row order need not match the XML order. At least two clusters are required. Review extra rows and the selected columns; the CSV format itself does not establish whether the clusters have a valid phylogenetic or sampling interpretation.

CSV-based clustered permutation: a single mapping permutes source-cluster identities across destination clusters. All taxa in a destination cluster draw original dates

from that same source cluster. Single-date source clusters therefore yield a common assigned date. For mixed-date source clusters, each recipient taxon independently samples a source observation with replacement; cluster means are not used. Some clusters may remain in place while others exchange. The original date configuration and duplicate configurations are excluded, and each accepted source-cluster mapping is used at most once.

Auto-cluster (Mclust on dates): fits a Gaussian mixture to the dates in the XML and then applies outside-cluster sampling adapted from Rieux & Khatchikian (2017). No CSV upload is needed. These groups describe the date distribution and need not correspond to monophyletic clades, locations, or transmission clusters.

Mclust settings: **Maximum number of clusters (G)** defaults to 9 (selectable 2–30 and capped by the number of taxa). BIC selects among G = 1 to that maximum. Model E assumes equal component variance and is the default; model V allows variance to differ.

Round dates to nearest integer before clustering is off by default and affects grouping only; dates assigned to randomized XMLs are still drawn from the original observations. At least five taxa and at least two inferred clusters are required.

Outside-cluster sampling: each taxon independently draws one date from observations outside its own inferred date cluster. Candidate pools are not depleted between taxa, so dates may be reused and their overall frequencies may change. This differs from CSV-based clustered permutation. Do not interpret the two strategies as interchangeable controls of temporal–genetic confounding.

Step 2: Set **Number of Randomizations** (default 10; requested range 5–20) and **File Prefix** (default DRT_rep), then click **Generate & Download ZIP**. The ZIP contains the generated XMLs and randomization_info.txt, which records the selected strategy, relevant clustering settings, and requested/actual replicate counts. If the finite CSV-based permutation limit is below the request, EvoTempo generates only the available number and displays a notification.

Before running BEAST, inspect the generated dates and output filenames. The generator adds replicate suffixes such as _01 and _02 to .log and .trees filenames. Templates with multiple loggers can still need manual filename checks within each replicate. Keep the original and randomized model settings and priors comparable, and use a separate output directory for each analysis strategy.

Note: the Mantel-test result does not automatically enable cluster-based DRT. CSV-based clustered permutation and automatic Mclust-based sampling use different randomization rules; select and report the intended strategy explicitly. Passing a DRT criterion does not guarantee that all sources of bias have been excluded.

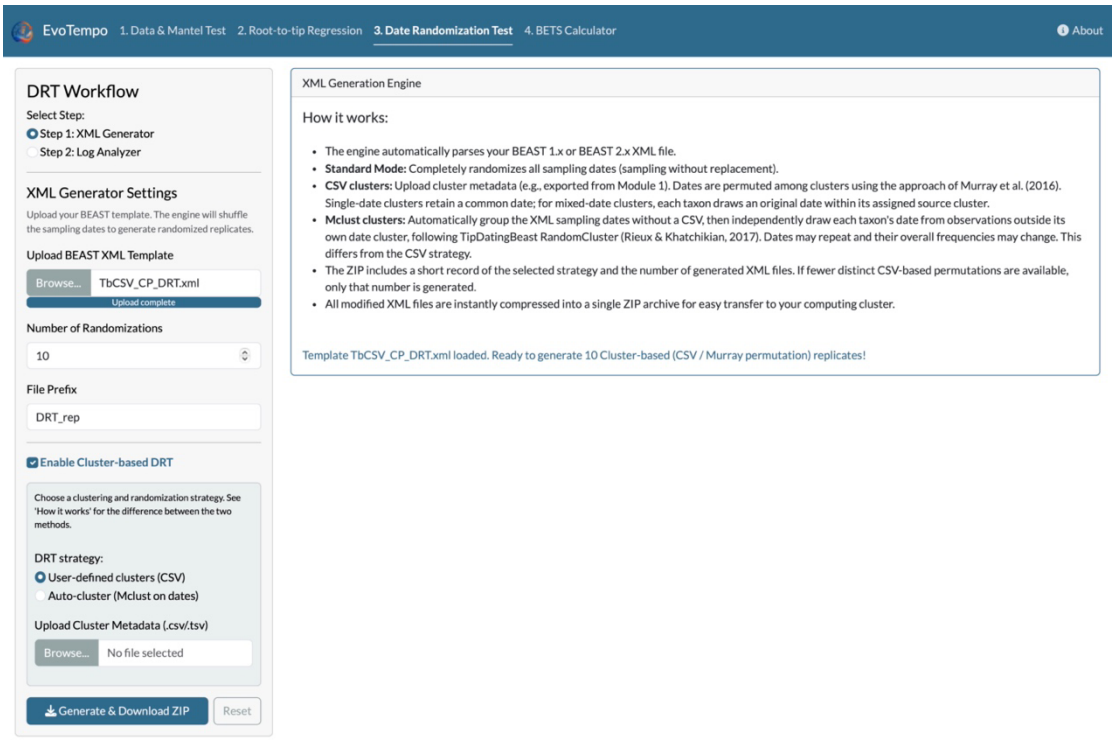


Figure 6 XML Generator in Module 3. The CSV and Mclust options select different clustering and randomization strategies, as described in Section 6.1.

6.2 Log Evaluator (CR1 & CR2 Criteria)

Once the BEAST runs are complete and their convergence has been checked, choose **Step 2: Log Analyzer.** Upload the **Original Log** and all **Randomized Logs** belonging to the selected strategy. Set **Burn-in (%)** (default 10; range 0–90). The first floor(n × burn-in fraction) rows are removed separately from each log. EvoTempo reports the

retained posterior mean and uses coda to calculate the 95% HPD interval for the selected parameter. It does not calculate effective sample sizes or check convergence.

Select the exact rate column in **Rate Parameter (Auto-detected)**. Common names include meanRate, clock.rate, clockRate, ucld.mean, and ucldMean; the names available depend on the XML and BEAST version. If needed, select or type the name manually, then click **Run Evaluation**. Every uploaded log must contain that same column; a missing column stops evaluation. In a relaxed-clock model, meanRate and ucld.mean can represent different summaries, so do not treat them as interchangeable aliases.

- **CR1** (Criterion 1, Standard): The posterior mean of the original rate estimate must fall outside the 95% HPD of every randomized replicate.
- **CR2** (Criterion 2, Stringent): The 95% HPD of the original estimate must show zero overlap with the 95% HPD of every randomized replicate.

Boundary conventions: both criteria use strict separation. A posterior mean exactly on a randomized HPD boundary fails CR1, and intervals that touch at an endpoint fail CR2. One overlapping replicate is sufficient for the corresponding criterion to fail. Interpret PASS or FAIL under the selected randomization design and model assumptions; neither outcome alone establishes the presence or absence of all temporal information.

The forest plot displays the original and randomized posterior means and 95% HPD intervals, with randomized entries ordered by their lower HPD bounds (**Figure 7**). **Export Plot Data** saves the summary as CSV, and **Export Vector PDF** saves the plot. Keep a separate record of the selected parameter and the uploaded log filenames when reporting the analysis.

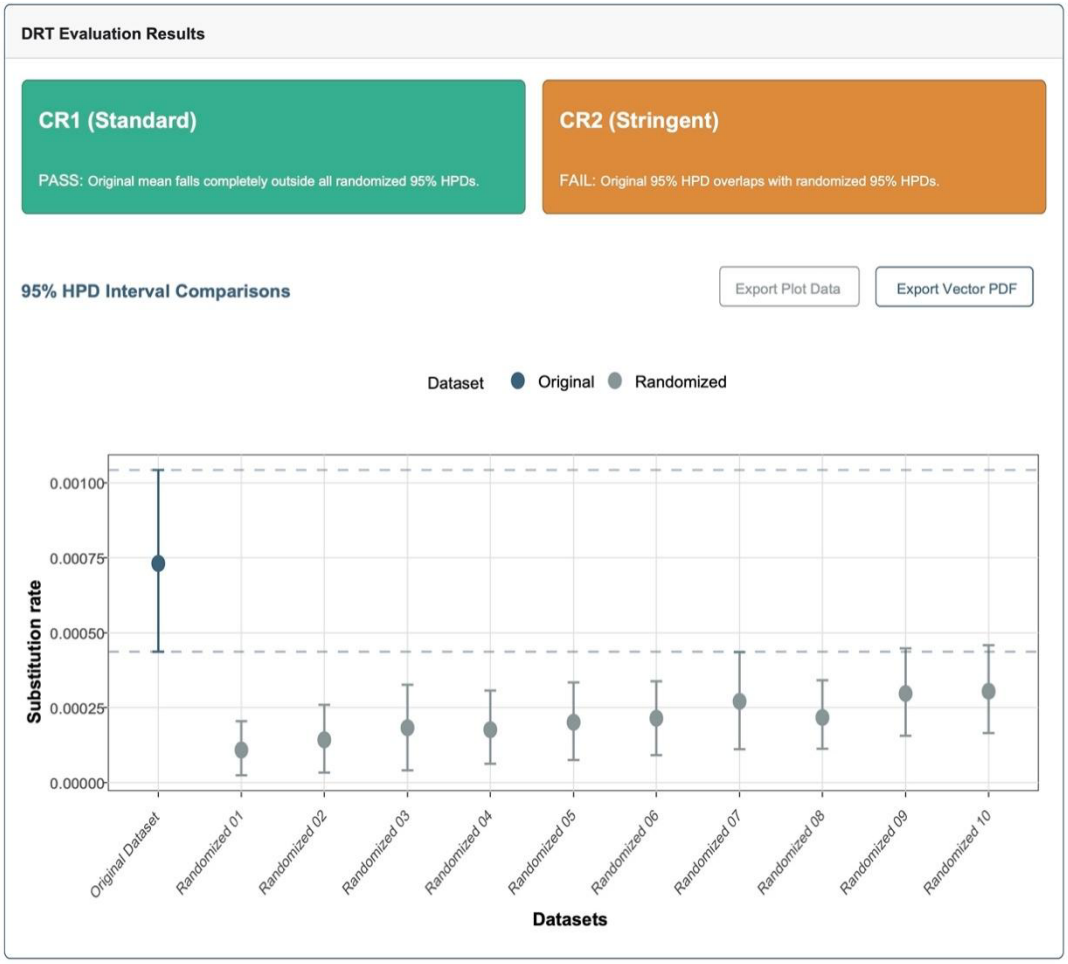


Figure 7 Posterior mean rates and 95% HPD intervals from the original and randomized analyses. CR1 and CR2 summarize their separation under the selected randomization design.

7. Module 4: Bayesian Evaluation of Temporal Signal (BETS)

BETS compares support for a model using the observed sampling dates with a corresponding model in which samples are contemporaneous. Module 4 is a calculator and visualization interface for externally estimated marginal likelihoods; it does not fit molecular-clock models, estimate marginal likelihoods, or select strict versus relaxed clocks.

7.1 Input Data

Estimate the marginal likelihoods of the two models externally in BEAST, using generalized stepping-stone (GSS), path sampling (PS), or stepping-stone (SS) sampling as appropriate for the configured models. Keep the compared model settings and priors comparable, use proper priors for marginal-likelihood estimation, and assess the reliability of the estimates before entering the results.

- **Heterochronous** model (M_{het}): uses the observed sampling dates to calibrate the analysis.
- **Isochronous** model (M_{iso}): constrains all sampling times to a common value while keeping the remaining model components comparable. The comparison evaluates the contribution of the sampling-date information under these model configurations.

Upload the final marginal-likelihood summaries, or enter the already calculated natural-log values manually. The reader searches the first 5,000 lines for a marginal-likelihood label, uses the last matching line, and extracts its first number. Always verify both extracted values: a step number or another numeric field may be read from a nonstandard summary. The interface label Log MLE refers here to log marginal likelihood, not a maximum-likelihood estimate.

7.2 Interpreting the Bayes Factor (BF)

EvoTempo calculates log(BF) by subtracting the log marginal likelihood of the isochronous model (M_{iso}) from that of the heterochronous model (M_{het}). Both inputs

must be on the natural-log scale and should not be log-transformed again. Positive values favor M_{het} , whereas negative values favor Miso. The displayed thresholds describe relative evidence on the natural-log scale, following Kass and Raftery (1995); they are not frequentist significance levels:

- log(BF) > 5: Very strong support for M_{het}.
- 3 < log(BF) ≤ 5: Strong support for M_{het}.
- 1 < log(BF) ≤ 3: Positive support for M_{het}.
- 0 < log(BF) ≤ 1: Weak or inconclusive support for M_{het}.
- log(BF) < 0: Greater support for Miso; the strength of that preference depends on the magnitude of log(BF).
- log(BF) = 0: The supplied log marginal likelihoods are equal; the comparison favors neither model.

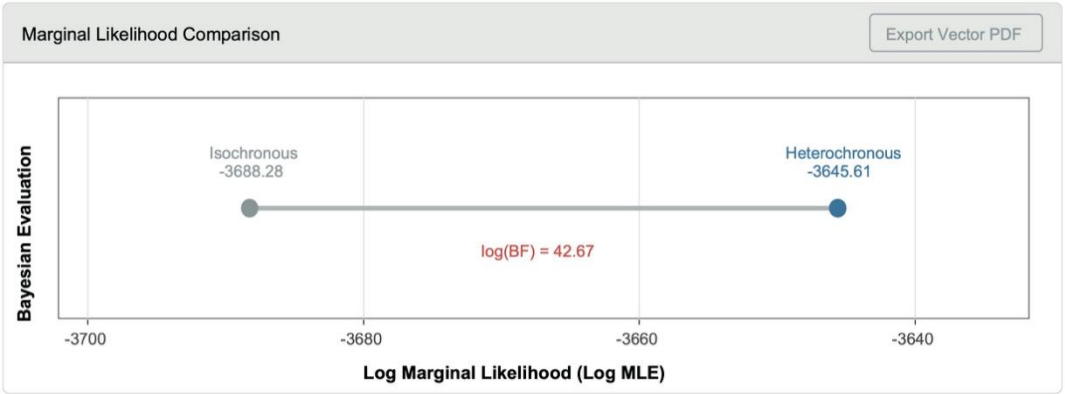


Figure 8 BETS dumbbell plot from Module 4, displaying log marginal likelihood estimates for competing heterochronous (M_{het}) and isochronous (M_{iso}) models.

7.3 Exporting the Log MLE Dumbbell Plot

Click **Evaluate Temporal Signal** to display the estimates and log(BF). The dumbbell plot compares the two models (**Figure 8**). Click **Export Vector PDF** to save it. Consider model adequacy, sampling design, and marginal-likelihood uncertainty when interpreting support.

8. Troubleshooting

- **Tree and RTT plot only show scatter points without connecting branches.**

Check that the uploaded tree has valid topology and interpretable branch lengths, then reload the data and recompute RTT. Confirm that the tree and date vector are present in Module 1. EvoTempo checks for missing branch lengths, but comprehensive validation of negative lengths is not implemented, so inspect the input rather than relying on an automatic warning.

- **Module 2 displays Waiting for Data.**

Load an alignment or tree and valid dates in Module 1, then click **Run Mantel Test** or **Send to RTT (Skip Mantel Test)**. Switch to Module 2 and click **Compute RTT**. Repeat the transfer and calculation after changing the source dates.

- **DRT Log Evaluator fails to read BEAST files.**

Inspect the header and select the exact rate column, including any BEAST 2 partition suffix. Use the same parameter in the original and every randomized log. Confirm that the files are posterior traces rather than marginal-likelihood summaries, and check for empty files or too few retained samples after burn-in.

- **BETS automatic extraction fails.**

Inspect the final estimator summary and enter its natural-log marginal likelihood manually if extraction is missing or incorrect. Some BEAST configurations write the summary to standard output. Use the same type of marginal-likelihood estimate for Mhet and Miso; an ordinary posterior log-likelihood trace is not a substitute.

- **ggtree installation fails via** install.packages()**.**

ggtree is distributed through Bioconductor. Install it using BiocManager::install("ggtree") and check R/Bioconductor compatibility. See Section 3.2 for the installation commands.

- **Package 'mclust' is required for auto-clustering** error in Module 3.

Install mclust with install.packages("mclust"), or choose **User-defined clusters (CSV)** when justified external cluster assignments are available. CSV-based clustered permutation and **Standard DRT** do not require mclust.

- **Mclust returns "only one cluster"**

Mclust selected one date cluster under the chosen model and candidate G values, so its outside-cluster sampling strategy cannot run. This does not establish that phylogenetic sampling structure is absent. Review the dates and settings, upload independently justified clusters through **User-defined clusters (CSV)**, or consider **Standard DRT** when appropriate.

- **CSV-based DRT rejects the cluster metadata.**

Check the selected taxon and cluster columns, duplicate or empty names, missing XML taxa, and missing cluster labels. At least two clusters are required. Cluster identifiers can be numeric or textual; they are normalized internally. Confirm that dates and groups represent the intended analysis before generating replicates.

- **Fewer CSV-based randomized XMLs are generated than requested.**

The finite number of distinct non-original clustered date configurations can be smaller than the requested count, especially with few clusters or repeated cluster dates. Check the notification and randomization_info.txt for the actual count. If no valid non-original configuration is available, or the generator cannot find the required distinct configurations, it stops with an error rather than writing invalid dates. Use the actual number of generated and successfully analyzed replicates when reporting the DRT.

9. References

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